

A convolutional neural network for acoustic-based intensity rainfall detection in the Amazon rainforest

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Abstract—Accurate rainfall monitoring is vital for agricultural planning and natural disaster prevention. This study introduces a novel convolutional neural network (CNN) architecture for passive acoustic-based rainfall detection in the infrastructure-scarce Amazon rainforest. For the binary wet/dry classification task, both the proposed and baseline architectures achieved metric values of approximately 99.00%, with the proposed model showing 36.14% fewer false alarms. Furthermore, on the multi-class rainfall-intensity task, the proposed CNN outperformed the baseline within the ambiguous moderate precipitation class, raising its recall rate from 48.90% to 60.70%.

Keywords—Acoustic, rainfall, deep learning, CNN.

I. INTRODUCTION

Monitoring rainfall remains a critical challenge in remote regions like the Amazon rainforest, where conventional instruments such as rain gauges (RGs) and satellite sensors are limited by dense vegetation, remoteness, and infrastructural constraints. Traditional RG networks in these areas are sparse and frequently non-operational, resulting in data gaps that hinder the hydrological and climate modeling essential for understanding regional precipitation dynamics. To overcome these limitations, reliable, low-cost alternatives are highly sought after [1], [2]. Environmental acoustic data, captured from the sound of raindrops impacting surfaces, has emerged as a viable solution due to its strong correlation with actual rainfall intensity [2].

Recent studies have demonstrated the utilization of acoustic data for rainfall detection, as shown in the following recent works. Wang et al. [1] proposed a novel rainfall observation method utilizing urban surveillance audio and CNNs. They introduced the RA_XZ dataset, evaluating 16 acoustic feature aggregation schemes across Mel spectrograms and Mel-frequency cepstral coefficients (MFCCs). Using 3-, 5-, and 7-stack CNN baseline architectures, their 7-stack model achieved 81.67% accuracy on RA_XZ and a state-of-the-art 93.38% on UrbanSound8k [3]. Notably, their 5-stack CNN achieved 92.90% accuracy on RA_XZ, slightly outperforming their 7-stack's 91.71% in that specific configuration, indicating

that a shallower model can offer high efficacy with lower computational complexity.

In [2], it is presented an innovative and low-cost method for estimating rainfall in the Amazon rainforest using sound recordings and machine learning (ML) models. By analyzing the acoustic signatures of rainfall, specifically the power spectral density (PSD) within a targeted frequency band, the researchers trained random forest classifiers and regressors on data collected from multiple sites. The models achieved accuracy of over 88% in rainfall detection and coefficient of determination (R^2) values exceeding 0.87 in estimating hourly rainfall rates, even at locations 30 km from the training site and across different vegetation types. The approach outperformed satellite-based estimates, accurately capturing total rainfall and offering high-resolution, real-time monitoring capabilities in remote, densely forested regions where traditional measurement systems are scarce. These findings, as well as in [1], highlight the potential of passive acoustic sensors combined with ML as a scalable solution for climate and hydrological monitoring in underrepresented tropical areas.

In this work, we propose a CNN structure based on the 5-stack architecture presented in [1]. The architecture suggested herein differs from [1] in three main aspects. First, we adopt a VGG-inspired design [4] that stacks small 3×3 convolutional with 2×2 max-pooling layers, in contrast to the larger receptive fields and strided convolutions of [1], while retaining its activation and regularization scheme (LeakyReLU, batch normalization and spatial dropout). Second, instead of training two independent networks, the same convolutional backbone is shared by the wet/dry and intensity models, which differ only in their deeper layers and output head, coupling representation learning across the two related tasks. Third, the intensity model is trained with a class-aware *focused mixup* [5], a targeted data-augmentation scheme that linearly interpolates only segments from the adjacent *light* and *moderate* classes, the boundary where confusion is most concentrated, leaving the well-separated *intense* class intact. As shown in Section IV, the proposed model achieves wet/dry detection that matches [1] and improves the rainfall-intensity classification, particularly in the difficult *moderate* class.

The remainder of this paper is organized as follows. A very short review on the composition of acoustic data related to rainfall is presented in Section II. The methods used in this paper are explained in Section III. We present the study area on Amazon forest, a brief data analysis, the methodology and the results in Section IV. Finally, the concluding remarks of this work are drawn in Section V.

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II. ACOUSTIC-BASED DATA FOR RAINFALL CLASSIFICATION

This section describes the log-Mel spectrogram (Mel) scale framework used to construct the acoustic images for the rainfall classification task. Widely adopted for feature extraction in audio pattern recognition [6], [1], the log-Mel representation converts the high-resolution discrete Fourier transform (DFT) spectrum into a compact set of perceptually inspired frequency bands followed by logarithmic amplitude compression. According to [6], this representation reduces the dimensionality and dynamic range of the spectral features while preserving the dominant spectral patterns, yielding a compact and robust input representation for audio recognition. The DFT maps each windowed segment to its complex spectrum $X(f)$; the (nonlinear) power spectrum $P(f) = |X(f)|^2$ is then mapped onto the Mel scale $M(f)$ as follows

$$M(f) = 2595 \log_{10} \left(1 + \frac{f}{700} \right). \quad (1)$$

After computing the log-Mel spectrograms for all training and testing samples, the resulting images are then saved at a resolution of 512×512 pixels. Fig. 1 illustrates the RGB spectrogram images for the no rain and intense rain classes, where distinct acoustic differences can be clearly observed between the two conditions.

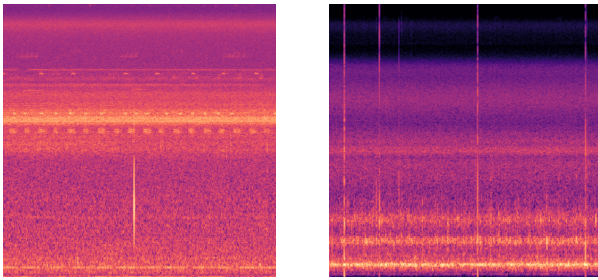


Fig. 1. Illustration of RGB-scale spectrogram images for the (a) no rain and (b) heavy rain classes.

III. PROPOSED CNN MODEL FOR RAINFALL DETECTION WITH ACOUSTIC-BASED DATA

A fundamental step in rainfall detection studies is to discriminate between wet (raining) and dry (not raining) periods, as well as their corresponding intensities light, moderate and intense [2] and, indeed, these are the main problems investigated in this work. As aforementioned, different methods can be applied to solve this problem, especially using deep learning (DL). In this work, a CNN architecture for rainfall detection is proposed to identify wet periods and determine its corresponding intensities.

A. Wet/dry discrimination

The wet/dry task is posed as a binary classification problem: given a short audio segment, the model decides whether it contains rainfall (*wet*) or not (*dry*). Each segment is represented as a single-channel (grayscale) log-Mel spectrogram resized to 256×256 , which is the input to the CNN depicted in Fig. 2

The architecture was initially derived from the rainfall audio classification CNN of Wang *et al.* [1], whose hyperparameters we adopted as a starting point.

Its architecture follows a VGG-inspired design [4], rather than the large receptive fields and strided convolutions of [1]. It stacks small 3×3 convolutional filters with a growing number of channels and 2×2 max-pooling. From [1] we retain the activation and regularization scheme, namely LeakyReLU ($\alpha = 0.1$), batch normalization and spatial dropout. Conv Blocks 1-3 use two convolutional layers with 32, 64, and 128 filters, respectively, followed by two single convolutional layers with 256 filters. The resulting feature maps are flattened and fed to a fully connected layer of 128 units, followed by a single sigmoid unit that outputs the probability of rainfall.

The model is trained with the Adam optimizer [7] (initial learning rate 10^{-3} , batch size 32) for up to 50 epochs, minimizing binary cross-entropy loss. The best model according to the validation loss is retained via model checkpointing and early stopping (patience 15), and the learning rate is reduced on plateau (factor 0.5, patience 5, minimum 10^{-6}).

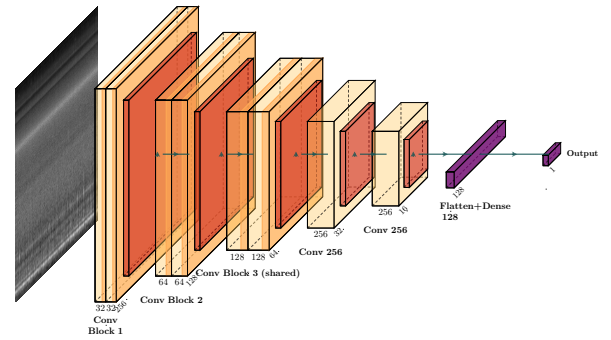


Fig. 2. Proposed CNN, shown for the wet/dry discrimination model. The rainfall-intensity model shares Convolutional Blocks 1-3 and differs only in the deeper layers and the output head.

B. Rainfall intensity discrimination

Once a segment has been classified as wet, the second task estimates its corresponding rainfall intensity. We follow the intensity rainfall bands of [8], [2] but group them into three levels *light*, *moderate* and *intense*, merging the heavy and violent bands into a single *intense* class. This is motivated by data availability, where violent events are rare in the recording, and keeping them as a separate class would yield a severely under-represented category and unstable training.

The multiclass model reuses the architecture of Fig. 2 and shares Conv Blocks 1-3 with the wet/dry model, differing only in its deeper layers and output head. The two deeper convolutional layers use 128 filters instead of 256. Following [1], the flattening stage is replaced by Global Average Pooling (GAP) [9], which yields a more compact and strongly regularized representation. The fully connected layer is reduced to 64 units, and the output is a three-way softmax. Training follows the same protocol as for the wet/dry model described in Section III-A, with two differences motivated by imbalanced classes: a batch size of 64 and a focal loss [10]

set of $\gamma = 2$ and $\alpha = [0.4, 0.4, 0.2]$, instead of binary cross-entropy, complemented by class-aware mixup augmentation, described as follows.

The mixup augmentation [5] constructs virtual training examples by linearly combining pairs of inputs and their one-hot labels. Given two segments (x_i, y_i) and (x_j, y_j) drawn from the same batch, a mixed sample $(\tilde{x}, \tilde{y})$ is produced as

$$\tilde{x} = \lambda x_i + (1 - \lambda) x_j, \quad \tilde{y} = \lambda y_i + (1 - \lambda) y_j, \quad (2)$$

with $\lambda \sim \text{Beta}(\alpha, \alpha)$.

We adopt a class-aware variant, hereafter referred to as *focused mixup*. The interpolation is applied only to pairs in which both segments belong to the adjacent *light* and *moderate* classes where most inter-class confusion concentrates while pairs involving the well separated *intense* class are left untouched. Mixing is triggered per batch with probability 0.5 and $\alpha = 0.4$, a setting that biases λ toward 0 or 1 and yields mixed samples close to the original training points rather than near the midpoint of the two classes.

Together with a higher dropout rate, this lighter head reflects the harder, fine-grained nature of the intensity task, where a heavier head tends to overfit the limited number of high-intensity examples.

IV. PERFORMANCE EVALUATION

In this section, we evaluate the performance of the two CNN architectures described Section III for both rainfall binary wet/dry and intensity discrimination tasks. First, in Section IV-A we provide a brief presentation of the dataset considered. Afterwards, the metrics used to compare the methods are introduced in Section IV-B. Then, details about the methodology used to evaluate each method are presented in Section IV-C while Section IV-D presents and discusses the obtained results. The code is available in <https://github.com/FabioBorgesA/A-CNN-for-acoustic-based-intensity-rainfall-detection-in-the-Amazon-rainforest>.

A. Adopted dataset

The authors collected an extensive dataset comprising approximately 799 hours of sound recordings captured by an autonomous Song Meter Micro recorder, sampling at 24 kHz, resulting in about 47,925 one-minute recordings recorded between February 14 and October 11, 2023. The sound data were collected in an open field area, located within the Mamirauá Institute for Sustainable Development (IDSM) campus in Tefé, Brazil. Ground-truth rainfall measurements were obtained using a tipping bucket RG (model pluviDB, DualBase), positioned about 15 meters from the recorder. The RG recorded rainfall intensities every 5 minutes with a precision of 0.2 mm, with rainfall categories assigned according to World Meteorological Organization criteria, including no rain, light, moderate, heavy, and violent rainfall. The IDSM recorder positioning is shown in Fig. 3.



Fig. 3. Placement of the sound recorder on Amazon area.

B. Evaluation metrics

The metrics used to evaluate the models are area under the receiver operating characteristic (ROC) curve (AUC), precision and recall [11].

The AUC is a scalar metric that summarizes the performance of a binary classifier across all possible classification thresholds by integrating the relationship between the false positive rate (FPR) and true positive rate (TPR) that forms the ROC [12]. To approximate the AUC [12] from a finite set of n discrete ROC points, the trapezoidal rule is applied as follows

$$\text{AUC} = \sum_{i=1}^{n-1} \frac{\text{TPR}_{i+1} + \text{TPR}_i}{2} \cdot (\text{FPR}_{i+1} - \text{FPR}_i). \quad (3)$$

Precision *Prec* quantifies how many cases are really positive classifications among all positive predictions. A value of 100% for it means that there were no false positives. It is given by

$$\text{Prec} = [\text{TP}/(\text{TP} + \text{FP})] \cdot 100\%, \quad (4)$$

where TP and FP, FN mean true positive and false positive, respectively.

Recall *Rec* measures how many cases are predicted as positive among all actual positive cases. A recall of 100% means that there were no false negatives. It is given by

$$\text{Rec} = [\text{TP}/(\text{TP} + \text{FN})] \cdot 100\%, \quad (5)$$

where FN means false negative.

C. Methodology

Each one-minute recording is sliced into eleven 10-second segments with a 5-second hop (50% overlap) [2]. Each segment is converted into a log-Mel spectrogram with 512 Mel bands using a short-time Fourier transform of size $n_{\text{FFT}} = 4096$ and a hop length of 512 samples, and the magnitudes are converted to decibels using the per-segment maximum as the reference level. Spectrograms are rendered as images with a magma colormap, saved at 512×512 pixels and resized to 256×256 before being fed to the network. The resulting image is used as a single-channel input for the wet/dry and rainfall-intensity model. During training of the intensity model, the input pipeline additionally applies SpecAugment [13]-style frequency and time masking together with mild brightness and contrast jitter, regularising the model against minor spectral and visual variations.

The dataset described in Section IV-A was partitioned at the audio-file level to prevent data leakage. Because consecutive

segments of a same recording share half of their content due to the temporal overlap, all eleven segments derived from the same one-minute audio are always assigned to the same split. A stratified 70/15/15 ratio was applied independently to each class, yielding training, validation and test subsets. For the wet/dry task, the binary test set comprises 4,752 segments (2,376 per class) corresponding to 432 audio files. For the intensity task, only segments labelled as wet are used, yielding 1,650 segments (550 per class) from 150 audio files balanced across *light*, *moderate* and *intense*.

Both the proposed CNN and the 5-stack baseline of Wang *et al.* [1] were trained and evaluated under identical conditions: same train/validation/test partitions, same input representation per task, and the same training protocol described in Section III. Model selection was performed via early stopping on the validation loss (patience of 15 epochs), and the checkpoint with the lowest validation loss was retained for final evaluation. All reported metrics are computed on the held-out test set, which was never used during training or model selection. Results are reported at two granularity levels: *per segment*, treating each 10-second spectrogram as an independent example; and *per audio*, aggregating the eleven segment-level predictions of each recording via hard majority voting over the per-segment argmax predictions into a single decision per minute of recording. The per-audio view reflects the operational unit of the system, in which a one-minute recording is classified as a whole.

D. Results

In what follows, we compare the proposed CNN against the 5-stack baseline of Wang *et al.* [1], reporting per-segment and per-recording precision and recall, as well as per-segment AUC, on the held-out test set for both the wet/dry detection and the rainfall-intensity classification tasks.

1) *Wet/dry detection*: At the segment level, both architectures discriminate wet from dry periods with very high fidelity, and their ROC curves are nearly superimposed Fig. 4, with an AUC of 99.31% for the proposed model and 99.36% for the baseline. As Fig. 5 shows, the proposed model raises 36.14% fewer false alarms over dry segments (53 FP from the proposed against 83 FP from the Wang). Otherwise, the confusion matrix (CMs) in Fig. 5 detail this asymmetry: the proposed model attains a recall of 97.8% on no-rain and 96.1% on rain, with 2.2% of dry segments misclassified as rain and 3.9% of rainy segments missed. The Wang baseline, in contrast, attains 96.5% and 98.9% recall, respectively, with a 3.5% FPR on no-rain and only 1.1% of rainy segments missed. Both models exceed 96% recall on each class, with their errors concentrated on opposite sides of the decision boundary.

When predictions are aggregated to the recording level via majority voting over the eleven segments of each recording, the precision/recall trade-off is preserved (precision 98.6% vs. 96.8%; recall 97.2% vs. 99.1%), confirming that the two designs differ in operating point rather than in detection capability.

2) *Rainfall-intensity classification*: The advantage of the proposed model is clearest on the three-class intensity task.

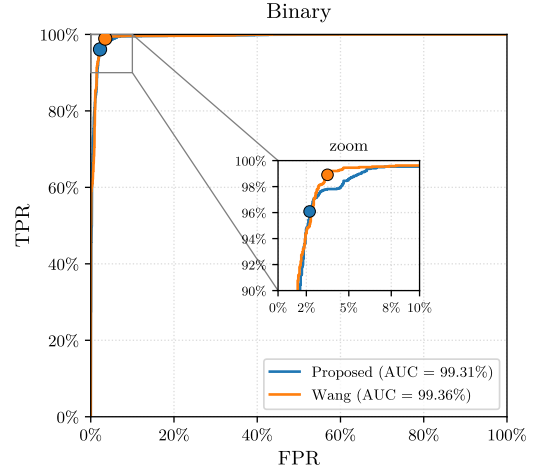


Fig. 4. ROC curves on the wet/dry task. Markers denote the operating points at threshold 0.5.

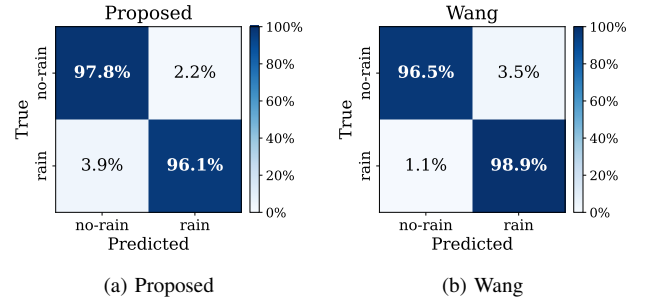


Fig. 5. Row-normalized CMs for the wet/dry task obtained with the (a) proposed and (b) Wang approaches.

The most prominent gain appears on the *moderate* class, where recall rises from 48.9% (Wang) to 60.7% (proposed) at the segment level, a direct reflection of the focused mixup augmentation, which targets exactly the *light*, *moderate* boundary. The CMs in Fig. 6 make this visible: when the baseline misclassified a *moderate* segment, 45% of those errors are absorbed by the adjacent *light* class, whereas the proposed model reduces this leakage to 33%.

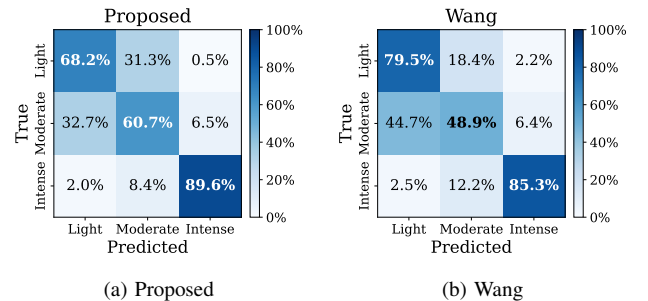


Fig. 6. Row-normalized CMs for the rainfall-intensity task obtained with the (a) proposed and (b) Wang approaches. Off-diagonal cells indicate inter-class confusion as a fraction of the true class.

As the Fig. 7 shows, the threshold-independent picture is consistent, with the proposed CNN attaining higher one-vs-rest AUC on *light* (86.8% vs. 85.6%) and *moderate* (78.5% vs. 77.3%), with the baseline marginally better on *intense*

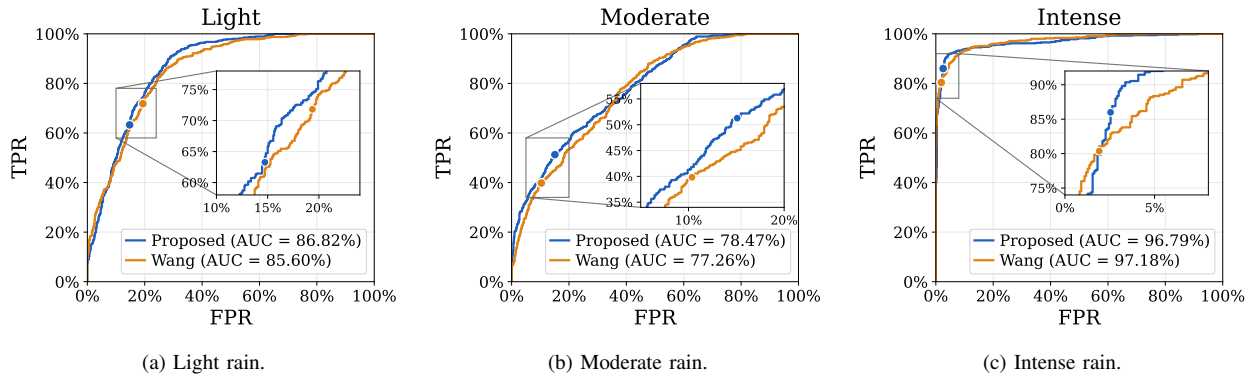


Fig. 7. Per-class ROC curves (one-vs-rest) for the rainfall-intensity task. From left to right: (a) light, (b) moderate, and (c) intense rain classes.

(97.2% vs. 96.8%). Aggregating predictions to the recording level preserves the same trends, with the *moderate*-class recall going from 48.0% to 64.0%.

The per-class behavior of precision and recall is summarized in Fig. 8. The proposed model gains on *moderate* (precision and recall) and on *intense* (both precision and recall) classes, while losing some recall on the *light* class, which the baseline tends to over-predict.

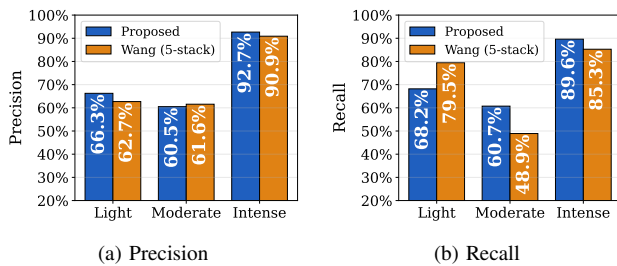


Fig. 8. Per-class (a) precision and (b) recall on the rainfall-intensity task. The proposed model improves over the baseline on the *moderate* and *intense* classes, at the cost of a lower recall on *light*.

3) *Discussion*: On wet/dry detection, the proposed CNN and the baseline are statistically equivalent in capability, their ROC curves overlap and the precision/recall trade-off persists across granularities with the proposed model favoring fewer false alarms. The contribution of the proposed design becomes substantive on the intensity task, where the combination of a lighter classification head, a single-channel input, and the focused mixup augmentation yields a consistent improvement on the ambiguous *moderate* class that dominates inter-class confusion, while also tightening the precision on the *intense* class.

V. CONCLUSIONS

In this work, the concept of leveraging acoustic data for rainfall monitoring in the Amazon rainforest was investigated using CNNs. A new CNN architecture was proposed and evaluated to identify wet and dry periods, as well as to classify rainfall intensity into light, moderate, and intense (heavy+violent) categories. The proposed model achieved wet/dry detection performance equivalent to the 5-stack baseline of Wang *et al.* [1], with AUC of 99.31% vs 99.36%, while producing 36% fewer false alarms on dry segments. The

primary contribution lies in the intensity task, the combination of shared convolutional backbone, per-image standardization, SpecAugment, and focused mixup yielded consistent gains over the baseline, with the most expressive improvement on the *moderate* class, whose recall rose from 48.90% to 60.70%. Directions for future work include investigating acoustic data-driven rainfall forecasting techniques following the initial classification tasks, as well as exploit potential multidimensional structures within the data.

REFERENCES

- [1] X. Wang, M. Wang, X. Liu, T. Glade, M. Chen, Y. Xie, H. Yuan, and Y. Chen, "Rainfall observation using surveillance audio," *Applied Acoustics*, vol. 186, p. 108478, 2022.
- [2] R. S. Xavier, M. Gosset, T. F. Maciel, T. Bicudo, L. A. d. Nascimento, E. Ramalho, and A. Fleischmann, "Measuring amazon rainfall intensity with sound recorders," *Geophysical Research Letters*, vol. 51, no. 20, 2024.
- [3] J. Salamon, C. Jacoby, and J. P. Bello, "A dataset and taxonomy for urban sound research," in *22nd ACM International Conference on Multimedia (ACM-MM'14)*, Orlando, FL, USA, Nov. 2014, pp. 1041–1044.
- [4] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. 3rd Int. Conf. on Learning Representations (ICLR)*, San Diego, CA, USA, 2015.
- [5] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "mixup: Beyond empirical risk minimization," in *International Conference on Learning Representations (ICLR)*, 2018.
- [6] S. Davis and P. Mermelstein, "Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 28, no. 4, pp. 357–366, 1980.
- [7] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. 3rd Int. Conf. Learn. Represent. (ICLR)*, San Diego, CA, USA, 2015.
- [8] C. Bedoya, C. Isaza, J. M. Daza, and J. D. López, "Automatic identification of rainfall in acoustic recordings," *Ecological Indicators*, vol. 75, pp. 95–100, 2017.
- [9] M. Lin, Q. Chen, and S. Yan, "Network in network," in *Proc. 2nd Int. Conf. Learn. Represent. (ICLR)*, Banff, AB, Canada, 2014.
- [10] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2017, pp. 2999–3007.
- [11] A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and Tensorflow: Concepts, Tools, and Techniques to Build Intelligent Systems*, 3rd ed. O'Reilly Media, Inc., 2022.
- [12] J. A. Hanley and B. J. McNeil, "The meaning and use of the area under a receiver operating characteristic (ROC) curve," *Radiology*, vol. 143, no. 1, pp. 29–36, 1982.
- [13] D. S. Park, W. Chan, Y. Zhang, C.-C. Chiu, B. Zoph, E. D. Cubuk, and Q. V. Le, "SpecAugment: A simple data augmentation method for automatic speech recognition," in *Proc. Interspeech*, 2019, pp. 2613–2617.